

Specific heat capacity (RP)

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1	State the three variables that a change in thermal energy depends on.	Mass, specific heat capacity and change in temperature	☐
2	What does $\Delta\theta$ represent?	Change in temperature	☐
3 ★	State the standard unit of specific heat capacity.	Joule per kilogram per degree Celsius (J/kg °C)	☐
4 ★	Define specific heat capacity.	The energy required to raise the temperature of 1 kg of a substance by 1 °C.	☐
5	A material has a high specific heat capacity. Is it hard or easy to heat up, and why?	Hard. It takes a lot of energy to raise the temperature of 1 kg by 1 °C.	☐
6 ★	State the three measurements needed to find the specific heat capacity of a substance.	Energy supplied, mass and temperature rise	☐
7	Name the piece of equipment used to measure temperature.	A thermometer	☐
8	Name the piece of equipment used to measure mass.	A balance	☐
9	Explain why a Bunsen burner is not used when measuring specific heat capacity.	The energy it supplies cannot be measured easily.	☐
10 ★	Explain why an electrical heater is used when measuring specific heat capacity.	The energy supplied can be calculated from potential difference, current and time.	☐
11	State the equation linking change in thermal energy, mass, specific heat capacity and temperature change.	change in thermal energy = mass $\times$ specific heat capacity $\times$ temperature change ($\Delta E = m c \Delta\theta$)	☐
12	When measuring specific heat capacity, explain why the block is wrapped in insulation.	To reduce energy transfer to the surroundings.	☐

	QUESTION	ANSWER	✓
13	Explain why a measured specific heat capacity is usually higher than the true value.	Some of the energy supplied is transferred to the surroundings, not the block.	☐